



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

IMPORTANT: This course is for the Double Major pathway only.

Programme	B.Sc. Physics Honours				
Course Title	SCIENCE COMMUNICATION				
Type of Course	Value-Added Course 2				
Semester	IV				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	3	-	-	45
Pre-requisites	Basic computer operating knowledge.				
Course Summary	This course introduces Latex programming for preparing scientific documents and presentations. This paper also introduces formal science communication, of which presentation and document writing forms a part.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Learn the basic structure of a LaTeX document, creating a new latex document	U	F	Instructor-created exams / Quiz / Practical Assignment
CO2	Understanding how to split a document into logical parts.	U	F	Instructor-created exams / Quiz / Practical Assignment
CO3	Understand text and paragraph formatting in Latex, including insertion of numbered and bulleted lists.	U	F	Instructor-created exams / Quiz / Practical Assignment
CO4	Understand how to insert tables, pictures, table of contents and equations in latex document.	U	F	Instructor-created exams / Quiz / Practical Assignment
CO5	Understand how to prepare a presentation using Latex.	U, Ap	F	Instructor-created exams / Quiz / Practical Assignment
CO6	Acquire the skillset required for formal science communication, including knowledge about journals, presentation skills and time management.	U	C	Instructor-created exams / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (36 +9)	Marks (50)
I	Document structure and basic commands		5	7
	1	Structure of a latex document: preamble, body/document environment. Creating a document by declaring its type: article, report, book, presentation (beamer), letter.	1	
	2	Options for the \documentclass command. The \title, \author, and \date commands. Creating abstract for articles: the abstract environment.	2	
	3	Starting and ending the body of our document: the \begin{document} and \end{document} commands. Splitting our document into segments: \part, \section, \subsection, \subsubsection commands. Creating numbered and non-numbered segments.	1	
	4	Optimizing space between words and sentences - quote-marks, slash marks, text mode superscript and subscript - dashes and hyphens, ellipsis, ready-made strings. font styles: bold, italic and underline text commands.	1	
Sections 5.1 - 5.3.4, 6.1, 6.3, 6.5, 6.7, 6.10, 6.12, 6.13 of Book 1				
II	Page, text & paragraph formatting		9	13
	5	Using Latex packages, two-sided documents, page dimensions, page size, margins, page orientation, margins, page size and rotation of a specific page, page styles, page background, multi-column pages, manual page formatting. (Chapter 16, Book1)	2	
	6	Paragraph formatting: paragraph alignment, paragraph indent and break, line spacing, manual breaks, verbatim text. Changing size of text, input encoding, escape codes. (Chapter 7, Book 1)	3	

	7	Less than ($<$) and greater than ($>$) symbols, degree symbol for temperature and math, other symbols in special environments. (Chapter 7, Book 1)	1	
	8	List Structures: itemize, enumerate, description environments. Nested lists, creating horizontal list using tasks package. (Chapter 10, Book 1).	3	
Sections 7.1 - 7.7, 10.1-10.3, 16.1 -16.13 of Book 1				
III	Inserting pictures and tables, mathematics		12	15
	9	Inserting table of contents.	1	
	10	Inserting pictures: The graphicx package, \includegraphics command, options for \includegraphics command: the scale, angle, options, supported image formats for compiling with pdflatex. The figure environment, captions for figures.	2	
	11	Mathematics environments, Symbols, Greek letters, Operators, Powers and indices, Fractions and Binomials, Roots, Sums and integrals, Brackets, braces and delimiters, Matrices and arrays, Adding text to equations, Formatting mathematics symbols, Colour	2	
	12	Plus and minus signs, Controlling horizontal spacing, dots in formulas, Equation numbering, Vertically aligning displayed mathematics, Indented Equations, Page breaks in math environments, Boxed Equations, Advanced formatting, Text in aligned math display, Changing font size.	2	
	13	Tables: The tabular environment, row specification, spanning, controlling table size, colors, width and stretching, table across several pages, partial vertical lines, vertically centred images, footnotes in tables, professional tables, sideways tables.	2	
	14	Presentations in Latex using Beamer: frames, title page, using presentation themes, frame customization, piece-wise presentation of slides, table and figure in presentation (21.1, 21.3.1, 21.4, 21.4.1,	3	

		21.5-21.5.6 of chapter 21, 22.1-22.1.4, 22.3 of chapter 22, Book 2). Dividing a Frame Column-Wise, Repeating Slides in Presentation, Numbering slides, Navigation buttons in beamer		
Sections 16.1 - 16.1.4, 21.1, 21.3.1, 21.4, 21.4.1, 21.5 - 21.5.6, 22.1 - 22.1.4, 22.3 - 22.5 of Book 2				
Sections 14.1 - 14.12, 17.1 - 17.10, 18.1 - 18.1.1, 27.1 - 27.22, 28.1 - 28.12, 41.1.5 , 41.1.10 of Book 1				
IV	Science communication		10	15
	15	Types of Science Communications- Research Publications, Conference Proceedings, Patents, Different Types Journals, The Process of Peer Review.	1	
	16	Quality Factors of a Journal, Subscribed Journals Versus Open Access Journals, Predatory Journals, Open Access to Scientific Knowledge, Popular Science Communication	2	
	17	Parts of a Research Paper: Writing the Introduction Section, Material and Methods, Experimental Methods, Results and Discussion.	2	
	18	Tables, Graphs, Images, Analysis, Justification, Validation, Limitation and Scope, Conclusion, Conflicts of Interest, References, Abstract, Ethics of Scientific Communication, Plagiarism.	2	
	19	Presentation Skills: Effective Oral Presentation, Norms for Preparing Slides and Presenting the Same, Converting a research paper to a presentation, Time Management in a Presentation.	3	
Relevant sections from Book 3 and Book 4				
V	Open Ended Module		9	
	Advanced beamer features, Designing of book			
	Sections from References: Relevant sections from Book 1 and Book 2			
Books and References:				

1. Latex, wikibooks. Free download from:
<https://upload.wikimedia.org/wikipedia/commons/2/2d/LaTeX.pdf> (Book 1)
2. LaTeX in 24 Hours: A Practical Guide for Scientific Writing, Dilip Datta, Springer 2017.(Book 2)
3. Effective Science Communication (Second Edition), Sam Illingworth and Grant Allen, IOP
4. Science Communication: A Practical Guide, MIT OpenCourseWare, John Durant and Bina Venkataraman
<https://ocw.mit.edu/courses/sts-034-science-communication-a-practical-guide-fall-2011/pages/lecture-notes/>

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	0	1	0	0	0	0	3	3	2	0	3	3	1
CO 2	0	1	0	0	0	0	3	3	2	0	3	3	1
CO 3	0	1	0	0	0	0	3	3	2	0	3	3	1
CO 4	0	1	0	0	0	0	3	3	2	0	3	3	1
CO 5	0	1	0	0	0	0	3	3	2	0	3	3	1
CO 6	0	1	0	0	0	0	3	3	2	0	3	3	1

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	